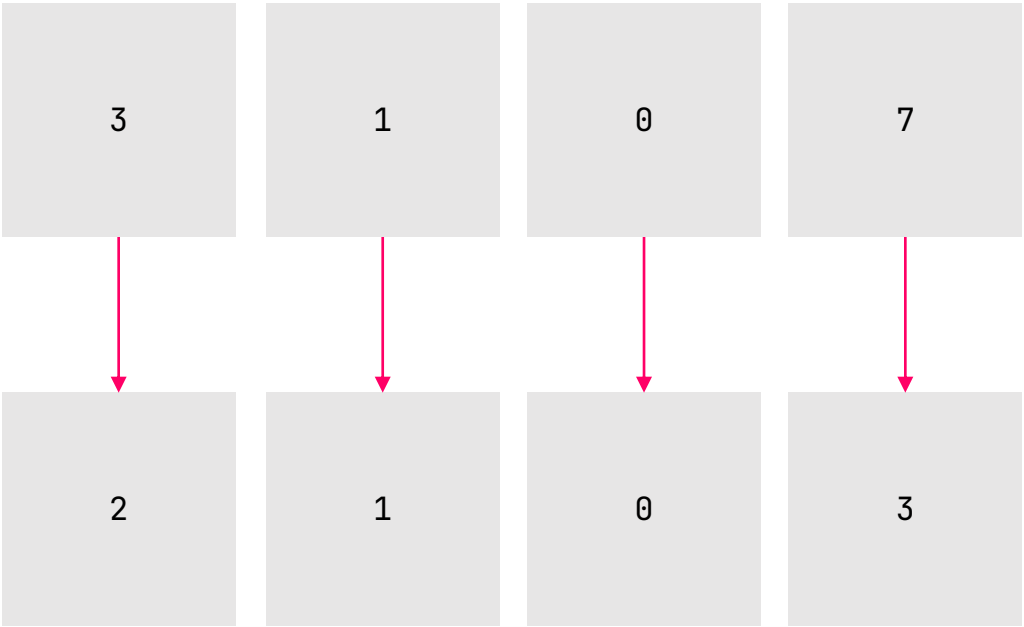
Coordinate Compression

Coordinate Compression

3	1	0	7

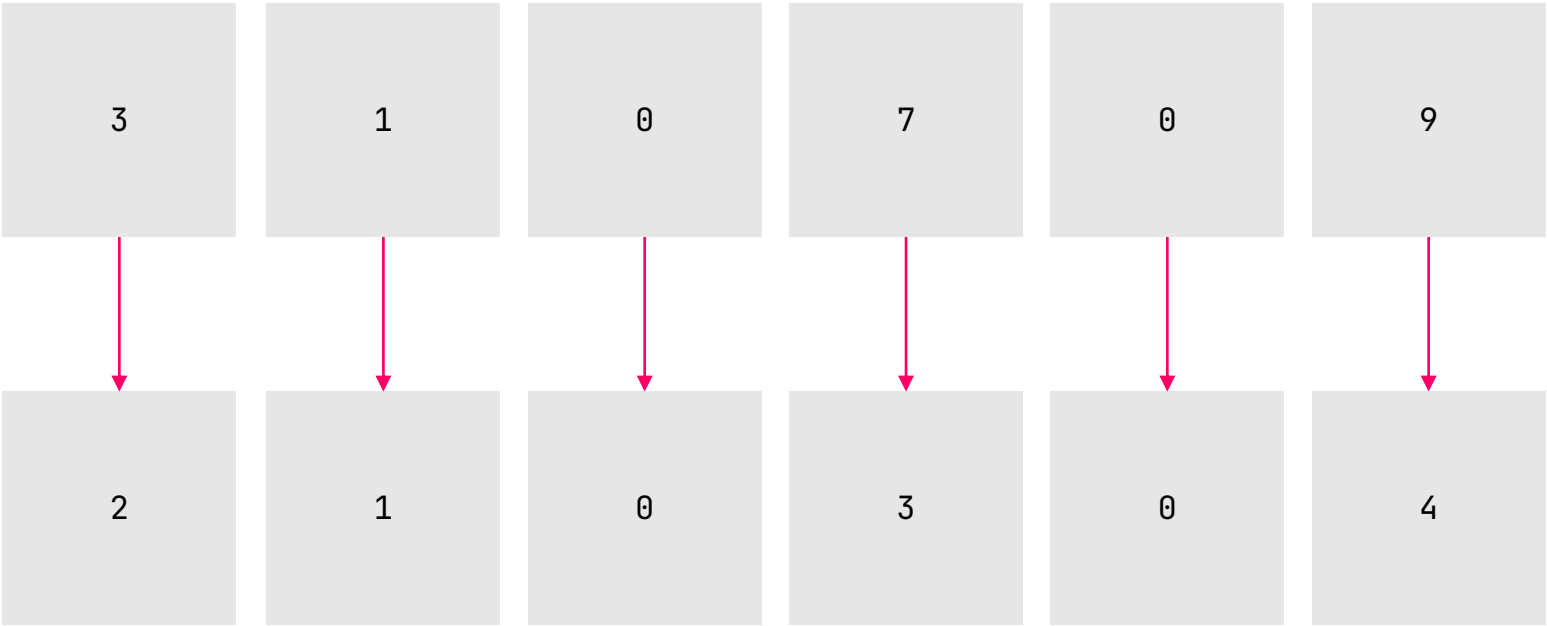
Coordinate Compression



Coordinate Compression

3	1	0	7	0	9

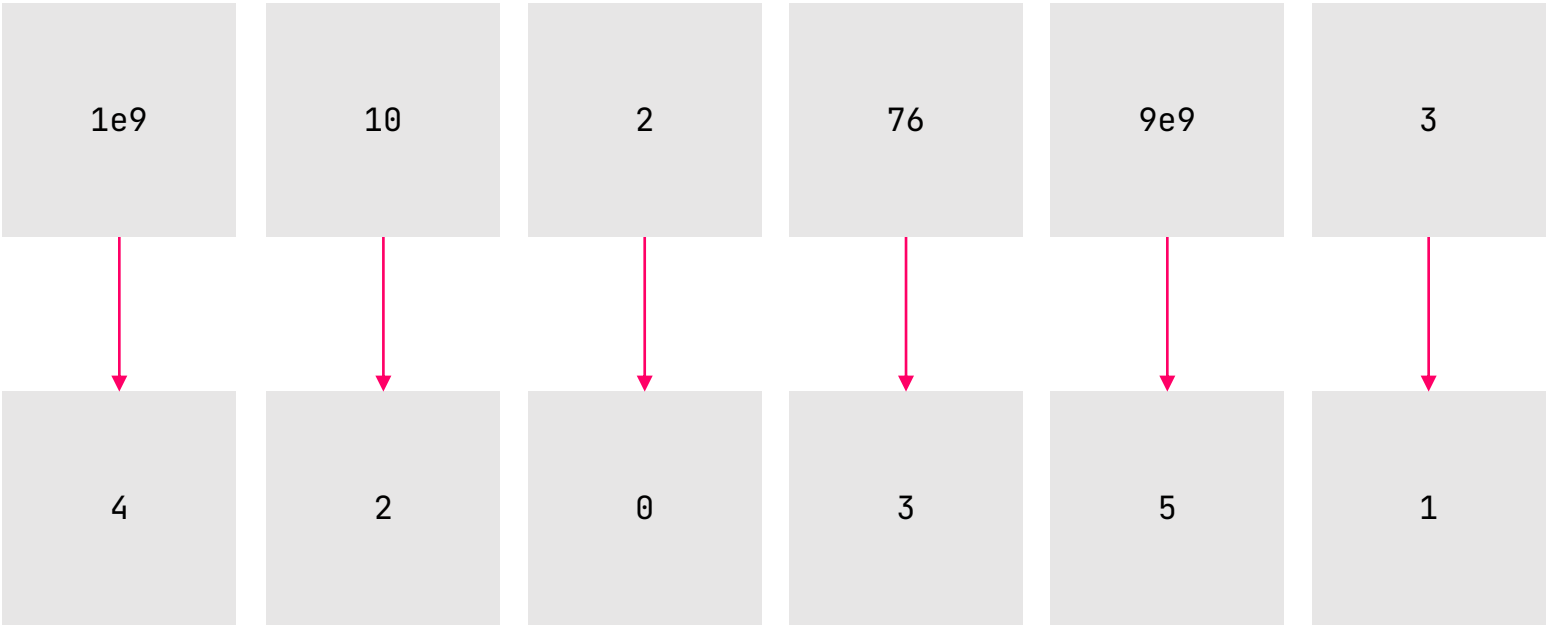
Coordinate Compression



Coordinate Compression

1e9	10	2	76	9e9	3

Coordinate Compression



Coordinate Compression

```
vector<int>v = {3 , 1 , 0 , 7 , 0 , 7} ;
vector<int>temp ;

void coordianteCompress () {
    temp = v ;
    // temp → {3 , 1 , 0 , 7 , 0 , 7}
    sort(temp.begin() , temp.end()) ;
    // temp → {0 , 0 , 1 , 3 , 7 , 7}
    temp.erase(unique(temp.begin() , temp.end()) , temp.end()) ;
    // temp → {0 , 1 , 3 , 7}
}

int getCompressedIndex(int val){
    return lower_bound(temp.begin() , temp.end() , val) - temp.begin() ;
    //return compressed index of value in vector v
}
```

Coordinate Compression

CSES

Restaurant Customers

Start

Coordinate Compression

You are given the arrival and leaving times of n customers in a restaurant.
What was the maximum number of customers in the restaurant at any time ?

Constraints

- $1 \leqslant n \leqslant 2 * 10^5$
- $1 \leqslant a < b \leqslant 10^9$

Coordinate Compression

Solution 1 : Brute Force

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	0	0	0	0	0	0	0	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 1 : Brute Force

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	0	0	0	1	1	1	0	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 1 : Brute Force

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	1	0	1	1	1	0	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 1 : Brute Force

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	1	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 1 : Brute Force O(n * max(b - a)) → (2 * 10^5 * 10^9)

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	1	0	0

Input :

3

5 8

2 4

3 9

The Maximum number of customers is 2

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	0	0	0	0	0	0	0	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	0	0	0	1	0	0	-1	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	0	-1	1	0	0	-1	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	1	-1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	1	-1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	1	-1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	1	-1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	-1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	1	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	0	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	0	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	-1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	1	-1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	1	0	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Solution 2 : Quick Sum

idx	0	1	2	3	4	5	6	7	8	9	10
timeline	0	0	1	2	1	2	2	2	1	0	0

Input :

3

5 8

2 4

3 9

The Maximum number of customers is 2

Coordinate Compression

Solution 2 : Quick Sum O(2n + max(b)) → O(n + max(b))

idx	0	1	2	3	4	5	6	…	10^7	…	10^9
timeline	0	0	0	0	0	0	0	0	0	0	0

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

Input :

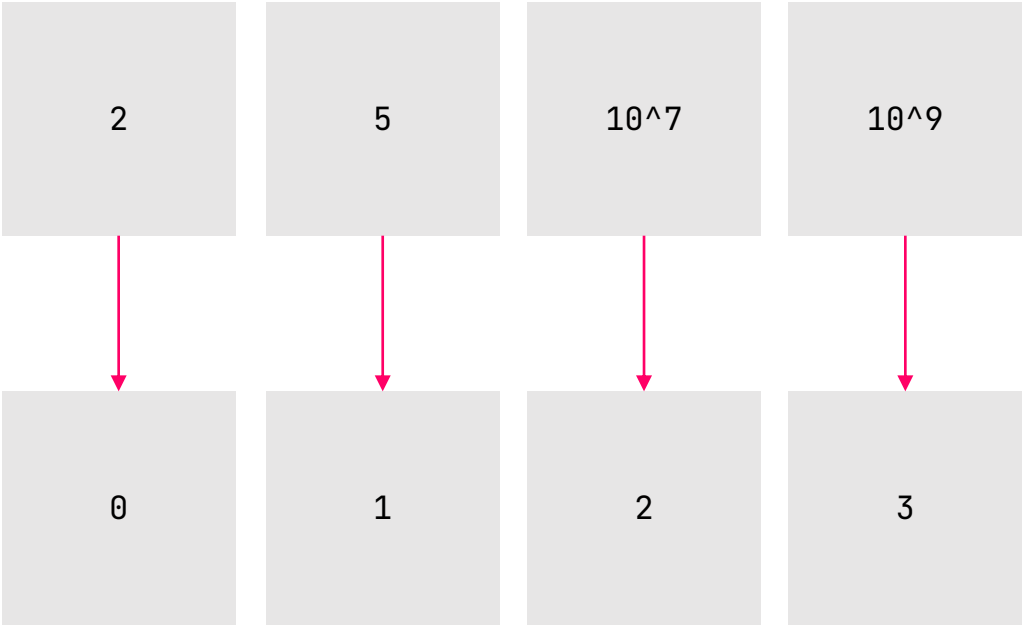
2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))



Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	0	0	0	0

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	0	1	0	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	1	-1	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	1	-1	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	1	-1	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	2	-1	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	2	1	-1

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	2	1	0

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	2	1	0

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

Input :

2

5 10^9

2 10^7

The Maximum number of customers is 2

Coordinate Compression

Solution 3 : Quick Sum + Coordinate Compression O(2n + 2nlog(2n)) → O(nlog(n))

idx	0	1	2	3
timeline	1	2	1	0

2	5	10^7	10^9
↓	↓	↓	↓
0	1	2	3

[2…4] → 1

[5…10^7-1] → 2

[10^7…10^9-1] → 1

[10^9…INF] → 0

Sweep Line

Sweep Line

Solution 4 : Sweep Line

idx	0	1	2	3	4	5
Events	2 , 1	3 , 1	4 , -1	5 , 1	8 , -1	9 , -1

Input :

3

5 8

2 4

3 9

Sweep Line

Solution 4 : Sweep Line

idx	0	1	2	3	4	5
Events	2 , 1	3 , 1	4 , -1	5 , 1	8 , -1	9 , -1
Customer	1	2	1	2	1	0

Input :

3

5 8

2 4

3 9

Coordinate Compression

Durian Company

[40 min]

Start

Coordinate Compression

1088_Iguana

[40 min]

Start

Coordinate Compression

CSES

Salary Queries

[60 min]

Start

Coordinate Compression

TOI17_Metaverse_2

[120 min]

Start